Operating Systems - Virtual Memory & Page Replacement

Course: Operating Systems | Semester 4 | Contributor: Alex Chen

Complete notes with diagrams on FIFO, LRU, Optimal page replacement, and TLB mechanisms.

Module 1: Virtual Memory Architecture

* Logical vs physical address translation through Multi-level Page Tables.
* Translation Lookaside Buffer TLB hit ratios and Effective Memory Access Time EMAT.
* Page faults handling sequence: interrupt, disk swap, frame allocation, page table update.

Module 2: Page Replacement Algorithms & Beladys Anomaly

* FIFO algorithm demonstration of Beladys Anomaly on 3 vs 4 frames.
* LRU approximation using reference bits, second-chance clock algorithm.
* Optimal Bˆ'lˆ¡dy page replacement as theoretical benchmark.

Module 3: Thrashing & Working Set Model

* Thrashing definition: system spends more time paging than executing instructions.
* Working-set strategy based on locality of reference parameter delta.

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